



CIRRHOSIS PATIENT DATA ANALYSIS

Patient Health Dashboard

Patient Health Executive
Overview

Demographic & Clinical
Analysis

Laboratory & Treatment
Analysis

Insight

Reccomondations

Death

161

Censored

232

Censore...

25

Status

▼ Enter Status →

Age_category

▼ Enter Age_category →

Gender

Fem...

Male

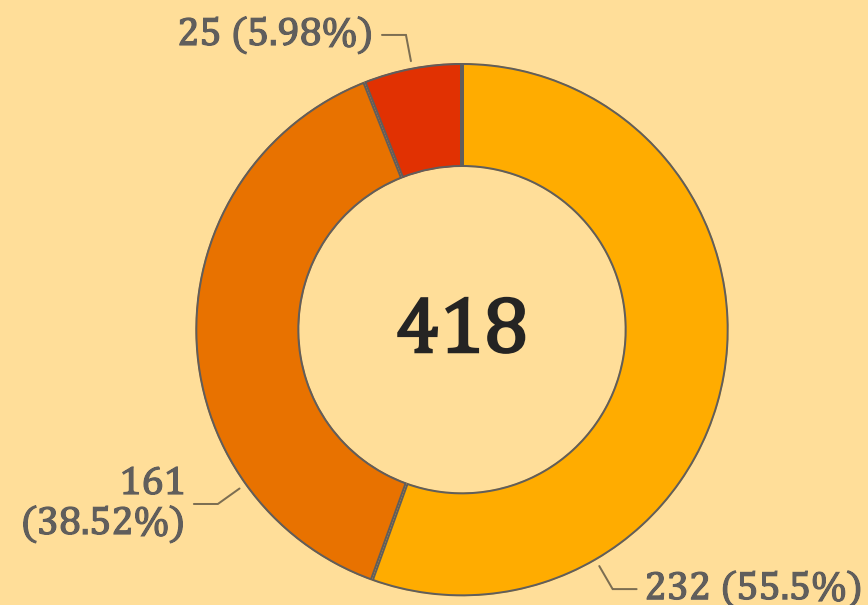
Stage

- ☐ 1
☐ 2
☐ 3
☐ 4

Drug

- ☐ D-penicillamine
☐ Placebo
☐ Unknown

Patient Status

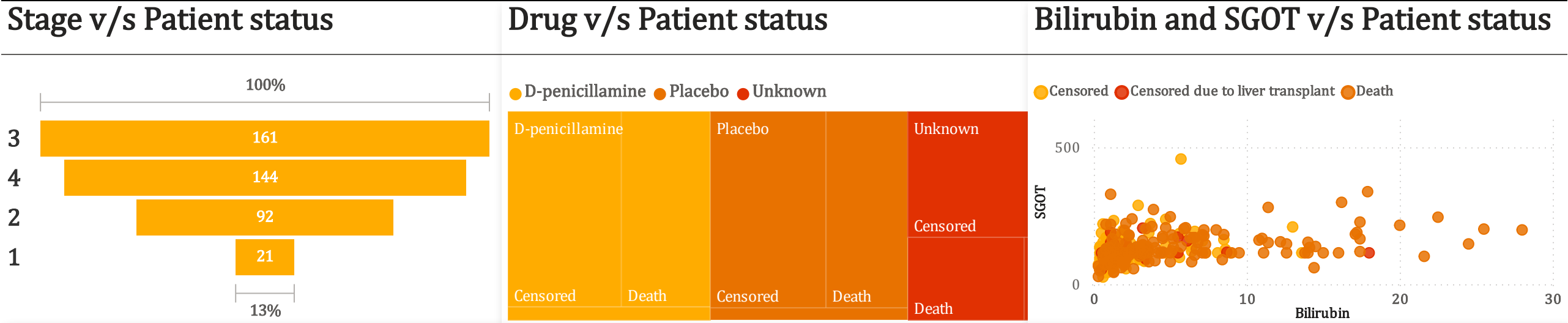


- Censored
- Death
- Censored due to liver t...

Patient Health Executive Overview	Demographic & Clinical Analysis	Laboratory & Treatment Analysis	Insight	Reccomondations			
51 Average age of patients							
Ascites Absent Present Unknown	Age_category v/s Status Censored Censored due to liver transplant Death Count of ID 200 0 Middle Adulthood Older Adulthood Young Adulthood Age_category 101 145 47 41 46	Gender v/s Patient status Censored Censored due to liver transplant Death Sex Female Male 0% 50% 100% Count of ID 57.49% 36.63% 38.64% 54.55%	Ascites v/s Patient status Absent Unknown Present 24 (5.74%) 106 (25.3...) 288 (68.9%)	Hepatomegaly Absent Present Unknown	Hepatomegaly v/s Patient status Censored Censored du... Death Hepatomegaly Present Absent Unknown 0 100 Count of ID 60 12 88 108 7 37 64 6 36	Spiders v/s Patient status Increase Decrease Total Count of Spiders 500 0 Censored Death Censored due to liver transplant Total 232 161 25 418	Edema v/s Patient status Censored Censored due t... Death Count of Status 354 216 116 44 20 No edema Edema prese... Edema refrac... to diu... Edema

LABORATORY & TREATMENT ANALYSIS

3.50 Average of Albumin	1.80K Average of Alk_Phos	3.22 Average of Bilirubin	350.27 Average of Cholesterol	91.28 Average of Copper	256.87 Average of Platelets	10.73 Average of Prothrom...	119.27 Average of Tryglicer...
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Laboratory measurements v/s Patient status

Status	Average of Bilirubin	Average of Cholesterol	Average of Albumin	Average of Copper	Average of Alk_Phos	Average of Tryglicer...
Death	5.54	384.74	3.36	121.07	2,295.77	
Censored due to liver transplant	3.56	403.10	3.49	111.76	1,468.92	
Censored	1.58	320.66	3.59	68.40	1,490.09	

Patient Health Executive Overview	Demographic & Clinical Analysis	Laboratory & Treatment Analysis	Insight	Reccomondations
KEY INSIGHTS				

★ Demographics

- Age: Middle adulthood has the highest deaths (101) followed by older adulthood (88).
- Sex: Females account for most deaths (137) compared to males (24) due to dataset imbalance.

★ Clinical Signs

- Ascites: Present ascites is seen in 24 deaths, indicating more advanced disease.
- Hepatomegaly: Hepatomegaly appears in 88 deaths, showing strong association with poor outcomes.
- Spiders: Spider angiomas occur in 52 deaths, marking visible liver deterioration.
- Edema: Refractory edema is linked to severe disease with 20 deaths, while controlled edema shows 26 deaths.

★ Treatment

- Drug: Mortality is similar between D-penicillamine (65 deaths) and placebo (60 deaths), showing no clear survival benefit.

★ Disease Stage

- Stage: Stage 4 has the highest mortality (84 deaths), followed by stage 3 (52 deaths).

★ Key Lab Indicators

- Bilirubin: Death cases show much higher bilirubin (5.54) than censored (1.57).
- Albumin: Albumin is lower in deaths (3.36) compared to censored (3.59).
- Alkaline Phosphatase: Death cases have significantly higher values (2295) than censored (1490).
- Copper: Copper is elevated in deaths (121) versus censored (68).
- Prothrombin Time: Higher in deaths (11.18) than censored (10.45).
- Platelets: Slightly lower in deaths (243) compared to censored (260).
- Cholesterol: Higher in deaths (385) than censored (320).
- SGOT: Elevated in deaths (136) compared to censored (109).
- Triglycerides: Higher in deaths (121) than censored (110).

RECOMMENDATIONS

★ Demographics

- Age: Prioritise early monitoring for middle-aged and older adults since they show the highest mortality.
- Sex: Ensure balanced sampling in future studies to avoid female-dominant datasets that may skew interpretation.

★ Clinical Signs

- Ascites: Strengthen routine screening and early management of ascites to prevent progression.
- Hepatomegaly: Treat hepatomegaly as a red-flag indicator and escalate care when present.
- Spiders: Use spider angiomas as supportive evidence of advancing liver disease and monitor closely.
- Edema: Pay special attention to refractory edema, as it signals severe disease requiring urgent intervention.

★ Treatment

- Drug: Since D-penicillamine and placebo show similar outcomes, consider reassessing treatment protocols and exploring alternative therapies.

★ Disease Stage

- Stage: Focus on early-stage detection and intervention to prevent progression to stage 4, where mortality is highest.

★ Key Lab Indicators

- Bilirubin: Use rising bilirubin as an early warning sign for deterioration and intensify monitoring.
- Albumin: Address low albumin promptly through nutritional and medical support to improve liver function.
- Alkaline Phosphatase: Treat high alkaline phosphatase as a marker of severe disease and investigate underlying causes.
- Copper: Monitor copper levels regularly, as elevated values indicate metabolic imbalance.
- Prothrombin Time: Use prolonged prothrombin time to identify patients at risk of bleeding complications.
- Platelets: Watch for declining platelets as a sign of portal hypertension or worsening liver function.
- Cholesterol: Consider high cholesterol as part of the metabolic profile of advanced disease.
- SGOT: Use elevated SGOT to track liver inflammation and adjust care accordingly.
- Triglycerides: Monitor triglycerides as part of overall metabolic assessment in severe cases.